PRACTICAL 4
CREATE USERS, ROLES, AND PERMISSIONS

Practical objective:
In this practical, you will understand Google Cloud Identity and Access Management, add a principal to a project, assign a limited role, verify access, create a service account, understand custom roles, and remove access after the practical.

Cloud platform used:
Google Cloud Platform, also called GCP.

Main service used:
Identity and Access Management, also called IAM.

Estimated outcome:
By the end of this practical, you will be able to:
- Open IAM in Google Cloud Console.
- Understand principals, roles, permissions, and allow policies.
- Add a user, group, or service account as a principal.
- Grant a predefined role using least privilege.
- Review assigned permissions.
- Create a service account for application identity.
- Understand basic, predefined, and custom roles.
- Remove temporary access after the practical.
- Review IAM configuration using current Google Cloud best practices.

Important safety message:
IAM controls access to cloud resources. Wrong IAM permissions can expose data, allow resource deletion, or create unexpected cost. For this practical, grant only limited test permissions and remove them after the lab.

1. BASIC THEORY BEFORE STARTING

1.1 What is IAM?

IAM stands for Identity and Access Management. It controls who can access Google Cloud resources and what actions they can perform.

IAM answers:
- Who is requesting access?
- Which resource is being accessed?
- What action is requested?
- Is the action allowed?

IAM is like the security permission system of Google Cloud. It decides who can view, create, modify, or delete resources.

1.2 What is a principal?

A principal is an identity that can be granted access.

Examples of principals:
- Google account user
- Google group
- Service account
- Cloud Identity domain
- Google Workspace domain

In older material, you may see the word "member." In current Google Cloud IAM, the word "principal" is commonly used.

1.3 What is a user?

A user is a human identity, usually represented by an email address.

Example:
student@example.com

Important note:
In Google Cloud IAM, you usually do not create normal Gmail users directly from IAM. You add existing users as principals or manage organization users through Google Workspace or Cloud Identity.

For this practical, "create user access" means adding an existing email identity to IAM with a specific role.

1.4 What is a role?

A role is a collection of permissions.

Example:
The Storage Object Viewer role contains permissions needed to view objects in Cloud Storage.

You normally assign roles, not individual permissions, to users or service accounts.

1.5 What is a permission?

A permission allows a specific action on a Google Cloud resource.

Examples:
- storage.objects.get
- compute.instances.create
- resourcemanager.projects.getIamPolicy

Permissions are very detailed. Roles group permissions together so access is easier to manage.

1.6 What is an IAM allow policy?

An allow policy is attached to a resource such as a project, folder, or organization. It contains role bindings.

A role binding connects:
- Principal
- Role
- Resource
- Optional condition

When you grant a role to a user on a project, you are adding a role binding to the project's IAM allow policy.

1.7 Types of IAM roles

Basic roles:
- Owner

- Editor
- Viewer

Predefined roles:
- Google-managed roles for specific services.
- Example: Storage Object Viewer, Compute Viewer, Logs Viewer.

Custom roles:
- User-defined roles containing selected permissions.

For modern cloud security, avoid basic roles for normal work. Prefer predefined roles, and use custom roles only when predefined roles are still too broad.

1.8 What is least privilege?

Least privilege means giving only the permissions required for a task and nothing more.

Example:
If a user only needs to view storage objects, grant Storage Object Viewer instead of Editor or Owner.

Least privilege reduces damage if an account is misused or compromised.

1.9 What is a service account?

A service account is a special identity used by applications, virtual machines, or automated processes.

Example:
A VM uses a service account to read files from a storage bucket.

Human users should use user accounts. Applications and services should use service accounts.

1.10 What is IAM condition?

An IAM condition restricts when a role binding applies.

Examples:
- Access expires after a date.
- Access applies only to a resource with a specific name.

Conditions are useful for temporary or limited access, but they should be used carefully because incorrect conditions can block needed access.

2. REQUIREMENTS

2.1 Hardware requirements

- Computer or laptop
- Internet connection

2.2 Software requirements

- Web browser
- Optional: Cloud Shell for command-line method

2.3 Cloud requirements

- Google Cloud account
- Google Cloud project
- Permission to manage IAM on the project
- A test user email or Google group email provided by the teacher, if role assignment is required

Required permission concept:
To manage access on a project, a user commonly needs a role such as Project IAM Admin or suitable custom permissions assigned by an administrator.

If you cannot add or remove IAM access, your account probably does not have IAM administration permission. Do not try to use another person's account.

3. STEP-BY-STEP PRACTICAL

STEP 1: Open Google Cloud Console

Open:

https://console.cloud.google.com

Sign in with your Google Cloud account.

Use your assigned lab account. IAM changes affect real access to cloud resources.

Expected result:
Google Cloud Console opens successfully.

STEP 2: Select the correct project

Use the project selector at the top of the console and select the project for this practical.

IAM roles can be assigned at project level. Selecting the wrong project may give access to the wrong resources.

Expected result:
The correct project is selected.

STEP 3: Open IAM and Admin

In the search bar, type:

IAM

Open:

IAM and Admin

IAM and Admin is where you view and manage access to the selected project.

Expected result:
IAM and Admin page opens.

STEP 4: Open the IAM page

In IAM and Admin, click:

IAM

The IAM page shows principals and their assigned roles on the selected project.

Expected result:
The IAM policy list is visible.

STEP 5: Observe existing principals

Look at the list of principals.

Observe:
- Principal name or email
- Principal type
- Assigned roles
- Inherited roles, if shown

Some roles may be inherited from folders or organizations. Inherited permissions come from higher levels in the resource hierarchy.

Expected result:
Existing access entries are identified.

STEP 6: Identify basic roles

Look for any roles named:
- Owner
- Editor
- Viewer

Basic roles are very broad. They are sometimes used in small labs, but production environments should prefer more specific predefined or custom roles.

Expected result:
You can identify whether basic roles exist in the project.

STEP 7: Plan the access to grant

For this practical, choose a limited predefined role.

Recommended lab role:

Storage Object Viewer

Purpose:
Allows viewing Cloud Storage objects without giving full project control.

Always decide the task first, then choose the smallest role that can perform that task.

Expected result:

The role to grant is selected.

STEP 8: Add a principal

Click:

Grant Access

or

Add principal

Enter the test user email or group email provided by your teacher.

Example:
student-test@example.com

In a real organization, granting roles to groups is often better than granting roles directly to individual users because group membership is easier to manage.

Expected result:
The principal email is entered.

STEP 9: Assign a predefined role

In the role field, search for:

Storage Object Viewer

Select:

Cloud Storage > Storage Object Viewer

This is a predefined role. Google manages predefined roles and updates them when services change.

Expected result:
The Storage Object Viewer role is selected.

STEP 10: Avoid basic roles

Do not select Owner, Editor, or Viewer for this lab unless your teacher explicitly instructs you.

Owner and Editor grant far more permissions than a beginner practical needs. Using narrow roles follows least privilege.

Expected result:
No broad basic role is selected for the test principal.

STEP 11: Optional condition for temporary access

If your teacher wants temporary access, add an IAM condition.

Example condition idea:

Access expires after a specific date and time.

IAM conditions can reduce risk by making access temporary, but they must be configured carefully.

Expected result:
Condition is added only if instructed.

STEP 12: Save the role binding

Click:

Save

Saving updates the project's IAM allow policy by adding the selected principal and role.

Expected result:
The principal appears in the IAM list with the selected role.

STEP 13: Verify the assigned role

Search or filter the IAM list for the principal email.

Confirm that the assigned role is:

Storage Object Viewer

Always verify IAM changes. A wrong role can give too much access or not enough access.

Expected result:
The correct role is visible for the principal.

STEP 14: Understand effective access

Check whether the principal has any other roles in the project or inherited from higher levels.

A user's real access is the combination of all roles granted directly and inherited from parent resources.

Expected result:
You understand that multiple role bindings can combine to create broader access.

STEP 15: Open Roles page

In IAM and Admin, open:

Roles

The Roles page shows basic, predefined, and custom roles available for the project or organization.

Expected result:

The Roles page opens.

STEP 16: Search for a predefined role

Search for:

Storage Object Viewer

Open or view the role details.

Role details show the permissions included in the role. This helps you understand what actions the role allows.

Expected result:
Permissions included in the predefined role are visible.

STEP 17: Observe permissions in the role

Read some permissions listed in the role.

Examples may include permissions related to reading storage objects and viewing bucket information.

Permissions are the actual allowed actions. Roles are collections of those permissions.

Expected result:
You can identify that the role contains multiple permissions.

STEP 18: Understand custom roles

On the Roles page, look for the option to create a custom role, but do not create one unless your teacher instructs you.

Custom roles are useful when predefined roles are too broad, but they require careful maintenance because permissions can change as services evolve.

Expected result:
You can identify where custom roles are managed.

STEP 19: Open Service Accounts

In IAM and Admin, open:

Service Accounts

Service accounts are identities for applications, VMs, and automation, not human users.

Expected result:
The Service Accounts page opens.

STEP 20: Create a service account

Click:

Create service account

Enter:

Service account name:
cloudnative-practical-sa

Service account ID:
cloudnative-practical-sa

Description:
Service account for Practical 4 IAM lab

Use clear names for service accounts so their purpose is easy to understand later.

Expected result:
Service account basic details are entered.

STEP 21: Grant limited role to service account

If asked to grant a role during creation, choose a limited role such as:

Storage Object Viewer

or skip role granting if your teacher wants only service account creation.

Do not give broad roles such as Editor to service accounts. Service accounts are often used by programs, so over-permissioning them is risky.

Expected result:
The service account is created with limited or no project role, depending on teacher instruction.

STEP 22: Avoid creating service account keys

Do not create or download a service account key file for this practical unless your teacher specifically instructs you.

Long-lived service account keys are sensitive credentials. Modern practice is to avoid keys when possible and use managed identity methods.

Expected result:
No unnecessary service account key is created.

STEP 23: Review service account details

Open the newly created service account and review:
- Email
- Description
- Permissions
- Keys tab

Service account email is used when attaching the service account to VMs,

applications, or workloads.

Expected result:
Service account information is reviewed.

STEP 24: Remove test user access

Go back to IAM page.

Find the test principal that was added.

Edit or remove the Storage Object Viewer role binding.

Temporary lab access should be removed after the practical. Keeping unnecessary permissions violates least privilege.

Expected result:
The test principal no longer has the lab role.

STEP 25: Delete the service account

Go to Service Accounts.

Select:

cloudnative-practical-sa

Click:

Delete

Confirm deletion.

Delete test service accounts after practicals so unused identities do not remain in the project.

Expected result:
The test service account is deleted.

STEP 26: Review IAM recommendations if available

Search for:

IAM Recommender

or check IAM recommendations in the console if available.

IAM Recommender can suggest removing or replacing overly broad roles based on permission usage history.

Expected result:
You can identify where IAM recommendations may appear.

STEP 27: Review Policy Analyzer if available

Search for:

Policy Analyzer

Open it if available in your account.

Policy Analyzer helps answer questions such as who has what access to which resources.

Expected result:
You can identify Policy Analyzer or understand its purpose.

STEP 28: Final access check

Return to IAM and confirm:
- Test user access was removed.
- Test service account was deleted.
- No broad role was granted accidentally.

Final IAM review is important because access mistakes can become security risks.

Expected result:
No unnecessary practical IAM access remains.

4. OPTIONAL COMMAND-LINE METHOD

This section is optional and should be used only if your teacher wants you to practice Cloud Shell.

4.1 Open Cloud Shell

Open Cloud Shell from Google Cloud Console.

Cloud Shell includes the gcloud CLI, which can manage IAM policies and service accounts.

4.2 Check active project

Run:

gcloud config get-value project

Always confirm the active project before changing IAM.

4.3 View IAM policy

Run:

gcloud projects get-iam-policy PROJECT_ID

Replace PROJECT_ID with your project ID.

4.4 Add a role binding

Only if instructed, run:

```
gcloud projects add-iam-policy-binding PROJECT_ID \
  --member="user:USER_EMAIL" \
  --role="roles/storage.objectViewer"
```

This grants Storage Object Viewer role to the user at project scope. Replace USER_EMAIL with the test email provided by your teacher.

4.5 Remove the role binding

Run:

```
gcloud projects remove-iam-policy-binding PROJECT_ID \
  --member="user:USER_EMAIL" \
  --role="roles/storage.objectViewer"
```

This removes the temporary role binding after the practical.

4.6 Create a service account

Run:

```
gcloud iam service-accounts create cloudnative-practical-sa \
  --description="Service account for Practical 4 IAM lab" \
  --display-name="cloudnative-practical-sa"
```

This creates an application identity. It does not automatically need broad permissions.

4.7 Delete the service account

Run:

```
gcloud iam service-accounts delete \
  cloudnative-practical-sa@PROJECT_ID.iam.gserviceaccount.com
```

Delete test service accounts after the lab to keep IAM clean.

5. OBSERVATION TABLE

Write your observations in this format:

Project ID:

IAM page opened:

Test principal email or group:

Role granted:

Was a basic role used?

Was the role predefined or custom?

Example permissions observed inside the role:

Was an IAM condition used?

Service account name:

Service account email:

Was a service account key created?

Was test user access removed?

Was the test service account deleted?

IAM Recommender or Policy Analyzer observed:

6. RESULT

A test principal was added to Google Cloud IAM with a limited predefined role. Role permissions were reviewed, a service account was created and inspected, and temporary access and test identities were removed after the practical.

7. CONCLUSION

This practical covered how Google Cloud IAM manages users, roles, permissions, and service accounts. They also learned the importance of least privilege, avoiding basic roles, using predefined roles, reviewing effective access, avoiding unnecessary service account keys, and removing temporary access.

8. LATEST-PRACTICES REVIEW TEST

This review checks whether the practical follows current recommended Google Cloud IAM practices.

1. Was the correct project selected before changing IAM?
   Yes. IAM changes must be made only in the intended project.

2. Was a limited predefined role used?
   Yes. A role such as Storage Object Viewer is better for this lab than Owner or Editor.

3. Were basic roles avoided?
   Yes. Basic roles such as Owner, Editor, and Viewer are broad and should be avoided for normal production access.

4. Was least privilege followed?
   Yes. The principal received only the access needed for the lab.

5. Was group-based access considered?
   Yes. In real organizations, groups are usually easier to manage than assigning roles directly to individual users.

6. Was the service account given limited or no role?
   Yes. Service accounts should not receive broad roles unless absolutely required.

7. Were service account keys avoided?
   Yes. Long-lived keys are sensitive and should be avoided when managed identity options are available.

8. Was temporary access removed after the practical?
   Yes. Temporary lab permissions should not remain after work is complete.

9. Was the test service account deleted?
   Yes. Unused identities should be removed.

10. Were IAM Recommender or Policy Analyzer concepts reviewed?
   Yes. These tools help audit access and move toward least privilege.

If the points above are covered, the practical aligns with current platform practices.
IAM practices and introduces current access-management recommendations.

9. VIVA QUESTIONS

1. What is IAM?
2. What is a principal?
3. What is a role?
4. What is a permission?
5. What is an IAM allow policy?
6. What are basic roles?
7. What are predefined roles?
8. What are custom roles?
9. What is least privilege?
10. What is a service account?
11. Why should service account keys be avoided?
12. What is IAM Recommender?
13. What is Policy Analyzer?
14. Why should temporary access be removed?

10. SHORT ANSWERS FOR VIVA

1. IAM controls who can access Google Cloud resources and what they can do.
2. A principal is an identity such as a user, group, or service account.
3. A role is a collection of permissions.
4. A permission allows a specific action on a resource.
5. An IAM allow policy contains role bindings that grant roles to principals.
6. Basic roles are broad roles such as Owner, Editor, and Viewer.
7. Predefined roles are Google-managed roles for specific services.
8. Custom roles are user-defined roles with selected permissions.
9. Least privilege means granting only the permissions required for a task.
10. A service account is an identity used by applications or services.
11. Service account keys should be avoided because they are long-lived sensitive credentials.
12. IAM Recommender suggests permission changes to reduce excessive access.
13. Policy Analyzer helps find who has what access to which resources.
14. Temporary access should be removed to reduce security risk.

11. COMMON MISTAKES TO AVOID

1. Granting Owner or Editor for simple tasks.
2. Giving access in the wrong project.
3. Granting roles directly to many users instead of groups.
4. Forgetting to remove temporary access.
5. Giving service accounts broad permissions.
6. Creating service account keys unnecessarily.
7. Ignoring inherited permissions.
8. Not checking role details before granting.
9. Assuming one role binding is the user's only access.
10. Not reviewing IAM permissions regularly.

12. PRACTICAL COMPLETION CHECKLIST

Before submitting this practical, confirm:

1. You opened Google Cloud Console.
2. You selected the correct project.
3. You opened IAM and Admin.
4. You viewed the IAM page.
5. You observed existing principals and roles.
6. You identified basic roles.
7. You selected a limited predefined role.
8. You added a test principal if instructed.
9. You assigned Storage Object Viewer or another limited role.
10. You avoided Owner and Editor roles.
11. You verified the assigned role.
12. You reviewed role permissions.
13. You opened the Roles page.
14. You understood custom role purpose.
15. You opened Service Accounts.
16. You created a test service account.
17. You avoided unnecessary service account keys.
18. You removed test user access.
19. You deleted the test service account.
20. You reviewed IAM Recommender or Policy Analyzer concept.
21. You completed the latest-practices review test.
22. You answered the viva questions.